

Final Project: ARIMAX model for a real-world time series

Course: Time Series Analysis

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Project objective

This project aims to apply the Box–Jenkins ARIMA methodology, including extensions for the treatment of calendar effects and outliers, to the analysis and forecasting of a selected real-world time series.

Students must choose one of the proposed monthly time series (or propose a different one for approval). The datasets are available in the ATENEA repository. **You must restrict your analysis until the year 2018.**

Important Information

- This assignment must be completed in pairs (two students per group).
- Submit the following files via the corresponding task on ATENEA:
 - o **Technical report file** (.pdf, .html, or .doc)
 - o **R script** (.R or .Rmd) containing the full analysis code
 - o **Analyzed dataset** (final dataset used in the analysis, including any transformations)
- **Deadline:** May 10, 2026 at 23:00.

Each group must also submit a **maximum 5-minute video**, with the participation of all group members, clearly explaining:

- the **purpose of the project** and the motivation for the chosen time series,
- the **problem or objective** being addressed,
- the **methodology applied**,
- the **main results obtained**,
- and the **key conclusions**.
- **Video requirements:**

Maximum duration: 5 minutes. Horizontal (landscape) format is required. Recommended format: .mp4 (720p or 1080p resolution). Audio must be clear and understandable. All group members must participate.

- The video may be recorded in **English, Spanish, or Catalan**.
- Upload the video to a cloud platform (Google Drive, OneDrive, or unlisted YouTube) and **submit the link via Atenea**.
- **ProMake** sure the **access permissions are correctly enabled**

Report Guidelines

- The report must not exceed **30 pages**, excluding appendices (e.g., additional code or extra figures).
- It should be written for a **technical audience**, not necessarily familiar with the specific time series. A brief contextualization of the dataset and its relevance must be included.
- The report must include at least the following sections:
 - o Table of contents
 - o Introduction
 - o Methodology (including classic ARIMA model development, calendar effects, and outlier treatment)
 - o Results and discussion
 - o Conclusions
 - o References (if applicable)
 - o Appendices (for supplementary material)
- Key charts and tables should appear in the main body of the report. Additional material may be included in the appendices.
- The R code must be clearly commented to facilitate understanding.

Final Project Requirements

A. Classical ARIMA Analysis

1. Identification

- Determine the transformations required to achieve stationarity. Justify your choices using both graphical and numerical evidence.
- Analyze the ACF and PACF of the stationary series to propose at least **two plausible models**. Clearly explain which features of the correlograms support your choices.

2. Estimation

- Estimate the candidate ARIMA models identified in the previous step (use the Arima function in R used in class).
- Evaluate model fit using appropriate metrics (log-likelihood, AIC, BIC, etc.).
- Compare models using AIC.
- Select the **two best-performing models** (lowest AIC values) to proceed with validation, forecasting, and ARIMAX analysis.

3. Validation: Are residuals white noise?

- Perform a complete residual analysis, justifying all assumptions using graphical tools and statistical tests.
- Analyze the infinite AR and MA representations, discussing causality and invertibility.
- Compare sample ACF/PACF with theoretical ACF/PACF.

4. Prediction Assessment

- Assess the stability of the selected models.

- Evaluate forecasting **accuracy** (RMSPE, MAPE) and **precision** (mean length of confidence intervals), reserving the last 12 observations for validation.

5. Forecasting

- Generate forecasts for the 12 months following the last available observation, including confidence intervals.
- Critically interpret the results and discuss model limitations in the context of the data.
- Provide the explicit statistical expressions of the selected ARIMA models.

B. ARIMAX Analysis

1. Outlier and Calendar Effects Treatment

For each of the two selected ARIMA models:

- Assess the significance of calendar effects.
- Apply automatic outlier detection and adjust the series accordingly. Interpret the identified outliers.
- Using the adjusted (linearized) series, repeat the full modeling process: identification, estimation, validation, and forecasting.

2. Results Comparison

- Compare the forecasting performance of ARIMA and ARIMAX models.
- Evaluate whether the inclusion of outliers and calendar effects improves model performance.

Report Submission Guidelines

- The report must include all steps and clearly justify all modeling decisions.
- It must be **self-contained**, meaning it should be understandable without reading the R code.
- Reports consisting only of compiled RMarkdown output without a proper structure will **not** be accepted.

Evaluation Criteria

Special consideration will be given to:

- Clarity of explanations
- Justification of decisions
- Ability to synthesize key insights

Conclusions

The conclusions section must include a **critical reflection** on the modeling process, including challenges encountered and potential improvements.

Additional Notes

- Extra credit will be given for incorporating **additional** statistical techniques beyond those covered in class.
- The submitted ZIP file must include the names of both students in the filename.

Final reminder: This is a **formal report** (including table of contents, introduction, methodology, etc.). A simple commented RMarkdown file is **not sufficient**.